



# Design, Analysis, and Test of a Cooled Rotating Detonation Combustor through axially injected stepped cooling slots

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**The present paper investigates the flow physics of a cooled rotating detonation combustor. First, an analytical model is developed for the slotted cooling designs, followed by a section on 3D RANS and URANS simulations to verify the analytical findings. Second, a cooling jacket is designed and machined to be integrated within an existing rotating detonation combustor. The jacket contains several measurements ports to measure temperature and pressure. Stators are added to measure stagnation pressure and achieve straight outflow. Finally, an experimental campaign is carried out on an existing RDE test platform with the axially injected stepped cooling slots. High speed imaging, as well as pressure and temperature probes are used to quantify the amount of coolant and the effectiveness of the cooling scheme.**

## 1. Introduction

Uncooled Rotating Detonation Combustors (RDCs) have been numerically analyzed and tested at universities, national labs, and industry all around the world. Most of these efforts have focused on detonation wave dynamics and uncooled RDC performance. There is still a dearth of research on the cooling of such combustors. Heat flux levels within RDCs were experimentally investigated by Theuerkauf et al. [1] Similarly, Roy et al. [2] performed conjugate heat transfer analyses by using periodic boundary conditions from Rotating Detonation Combustor simulations. Braun et al. [3] investigated the convective heat flux within RDCs and found that the average convective heat flux levels are nearly the same between the detonation front and downstream of the triple point. At the National Energy Technology Laboratory [4], a water-cooled RDC has been tested recently. At Nagoya University, hollow combustors are cooled through small holes on the outer wall [5]. The objective of the present research is to numerically analyze, and experimentally validate the performance of an RDC with hydrogen-air combustion, stepped axial injection cooling slots, and nitrogen gas film cooling.

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## 2. Numerical Analysis of the Cooled Combustor

### 2.1. Heat Flux Estimation

To estimate the cooling required for the combustor, a simulation of an uncooled rotating detonation combustor was used. The simulations were performed with CFD++ of Metacomp [6] on a structured mesh with about 45 million cells, which was used and validated in previous work[7,10]. The combustor is a non-premixed combustor running on a hydrogen-air mixture [7]. The viscous sublayer was resolved to ensure capturing of the viscous sublayer. The wall temperature was set to 600K. Figure 1[7]. The viscous sublayer was resolved to ensure capturing of the viscous sublayer. The wall temperature was set to 600K. Figure 1a and b depict the instantaneous heat flux on the wall in which the highest heat fluxes are noted downstream of the detonation wave. Figure 1c depicts the average heat flux along the axial direction, with average heat loads up to 3 MW/m<sup>2</sup>.

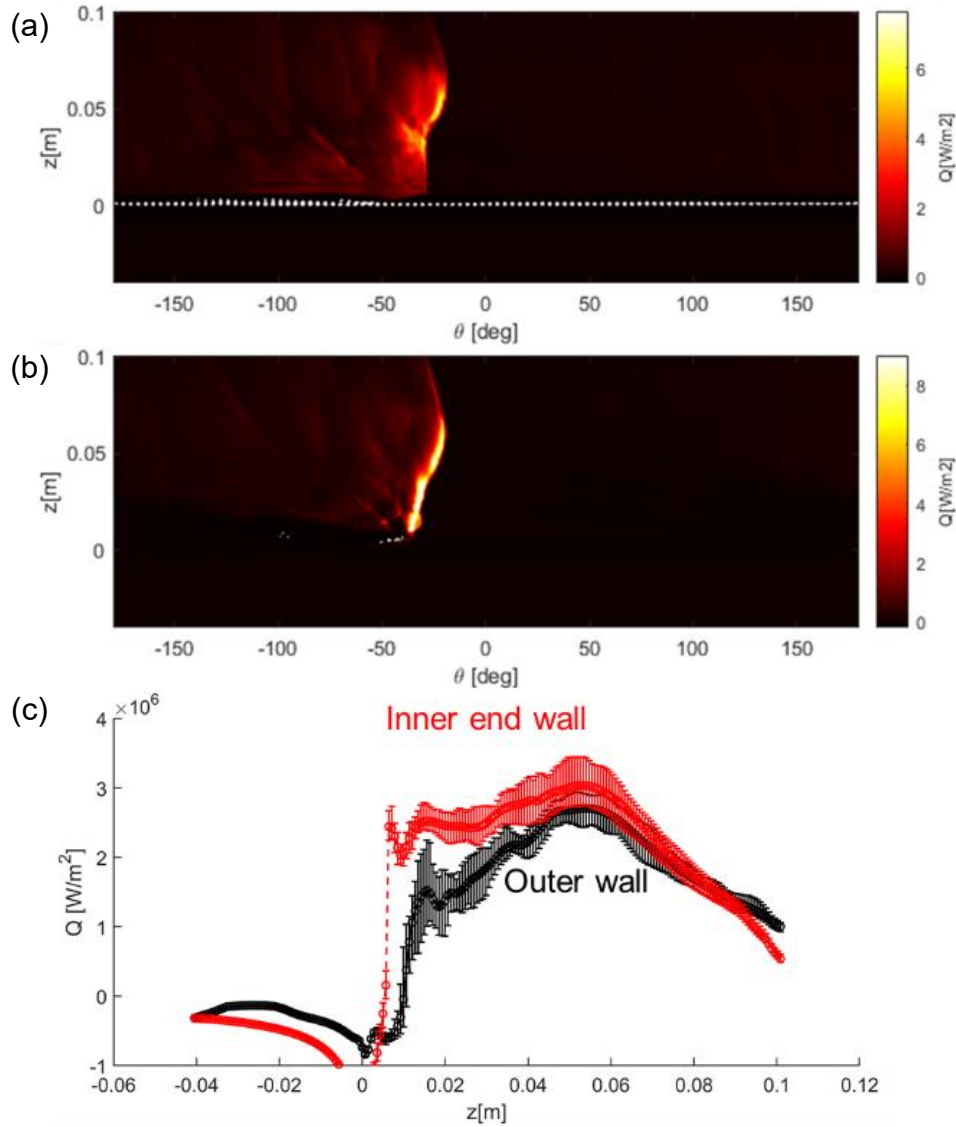


Figure 1: Heat flux from the uncooled THOR combustor at 1.2 kg/s: (a) instantaneous snapshot on the inner end wall, (b) instantaneous snapshot on the outer end wall, (c) average heat flux along the axial direction.

## 2.2. Preliminary Design Choices for the Cooling Jacket

Due to the high levels of wall heat flux, as noted above, rotating detonation combustors require a cooling scheme to allow continuous operation. One possible solution is the use of gaseous film cooling to protect the walls of the combustion chamber. While this cooling scheme is pervasive in the deflagration combustors of turbomachine engines, additional challenges are faced when designing a cooling setup for a rotating detonation combustor. The fluctuating pressure levels in the combustion chamber mean that coolant flow rates will fluctuate as well. As the moving detonation wave passes an injection point, the increase in pressure could even lead to momentary ingestion of hot air into the coolant reservoir. To mitigate the effects of the variable combustion chamber pressure, coolant pressure could be increased to choke the cooling passages. Choking the passages would make the coolant flow rate insensitive to the changing pressures in the combustion chamber. However, increasing the coolant pressure also leads to increased coolant blowing ratio, which poses additional challenges in the design of the film cooling setup. In his summary of advancements in film cooling, Goldstein [8] reviewed the results of many previous researchers. These studies showed that for different film cooling layouts, high blowing ratios can have either a positive or negative effect on cooling effectiveness. For those layouts with coolant injection perpendicular to the mainstream, coolant jets with high blowing ratios will blow off the surface and thus not form the desired film of cool air. For layouts with injection parallel to the mainstream flow direction, higher blowing ratios lead to increased film cooling effectiveness.

In addition to the choice of injection angle, designers also have a choice between injection through either discrete holes or continuous slots. As shown in Goldstein's review, injection through discrete holes can lead to hot spots between requiring additional coolant mass flow to prevent exceeding of wall temperature limits. Continuous slots avoid this behavior.

For the aforementioned reasons, the preliminary cooling design focused on injection tangential to the surface using continuous slots. Goldstein describes a variety of models for film cooling effectiveness with this setup [8]. For slot tangential injection, the primary two are

$$\eta = \frac{1}{1 + \left[ 0.329 \frac{c_{p\infty}}{c_{p2}} \xi^{0.8} \right]} \quad (1)$$

$$\eta = \frac{1}{1 + \frac{c_{p\infty}}{c_{p2}} [0.329(4.01 + \xi)^{0.8} - 1]} \quad (2)$$

Where,

$$\xi = \frac{x}{M_s} \left[ \frac{\mu_2}{\mu_\infty} Re_2 \right]^{-0.25} \quad (3)$$

Between these two equations, equation (1) assumes that the boundary layer starts at the point of injection while the equation (2) assumes that the boundary layer has started upstream of the point of injection. Equation (1) gives higher adiabatic wall temperature estimates, so it was selected for the analysis. Wall thickness was assumed to be 1.5 mm. The limit on wall thickness was to aid in manufacture and to make the wall more able to withstand the high pressures of the passing wave. From Goldstein, having a thick wall compared to the slot height reduces the cooling effectiveness, so the slot height was set at 1.5 mm to limit the detriment in performance. From the plots in Goldstein's work, this will still come with around a 15% reduction in cooling effectiveness, increasing slot height further would increase slot area further and require higher cooling mass flows.

Maximum coolant pressure was assumed to be four bar while the minimum pressure was determined from the pressure ratio required to keep the slot choked at the average combustor pressure. Figure 2 shows the results of a 1-D analysis varying coolant pressure and number of slots. Figure 2a shows that increasing the number of slots and coolant pressure increases mass flow while decreasing maximum adiabatic wall temperature. There are diminishing returns

with respect to both pressure and slot count. Figure 2b shows a trace of the expected adiabatic wall temperature for a 1 and 3 slot design with a coolant pressure of 3 bar.

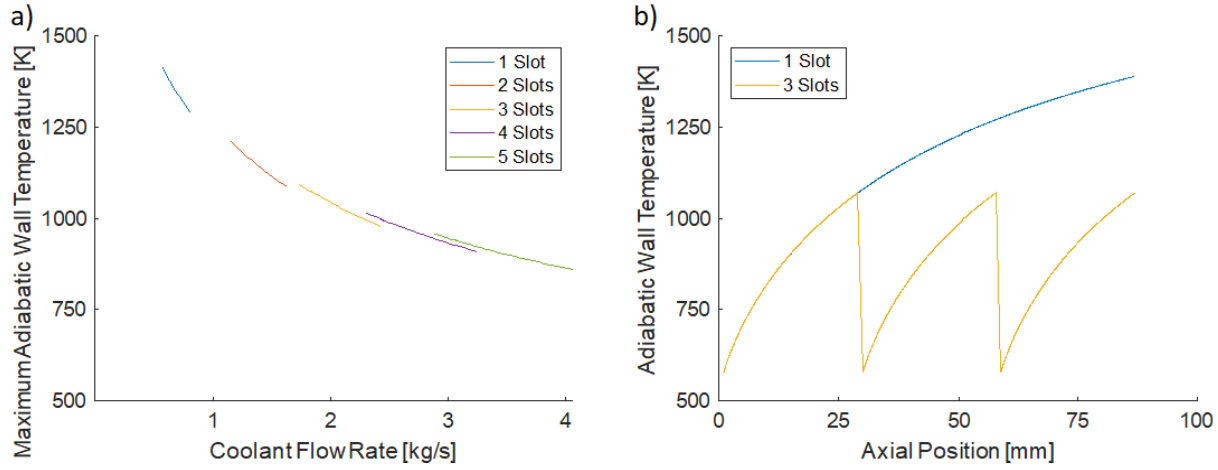


Figure 2: a) Impact of variable coolant pressure on mass flow and maximum wall temperature, b) wall temperature profiles for 1 and 3 slot designs at a pressure of 3 bar.

This analysis is an upper estimate of wall temperature. The less optimistic correlation for film cooling effectiveness was used. It only considers the cooling effectiveness from the most recent slot and ignores that upstream coolant flows are either mixed into the mainstream or stay in a coolant layer past the next slot. In addition, it ignores that the flow of coolant would be cooling the other side of the wall as well. For these reasons, and considering a maximum wall temperature of 1000 K, a three-slot design was chosen for further analysis.

### 2.3. URANS Calculations

As described in 2.2, a cooling mechanism with a three continuous concentric slots in a stepped combustor wall was selected for this RDC.. A 2D representation of the fluid domain is given in Figure 2a. The computational mesh was made of a structured mesh in ICEM. The total mesh size was around 50 million cells (Figure 2b), in which care was taken to resolve the viscous sublayer in the main combustion chamber. To simplify the geometry, a premixed simulation was performed, in which hydrogen-air was imposed at the inlet of the domain. The first two slots require higher feed pressures than the third slot which lies in the transonic/supersonic exhaust of the combustor. Additionally, downstream of the slots, a diverging section allows the combustor to fully expand to supersonic conditions.

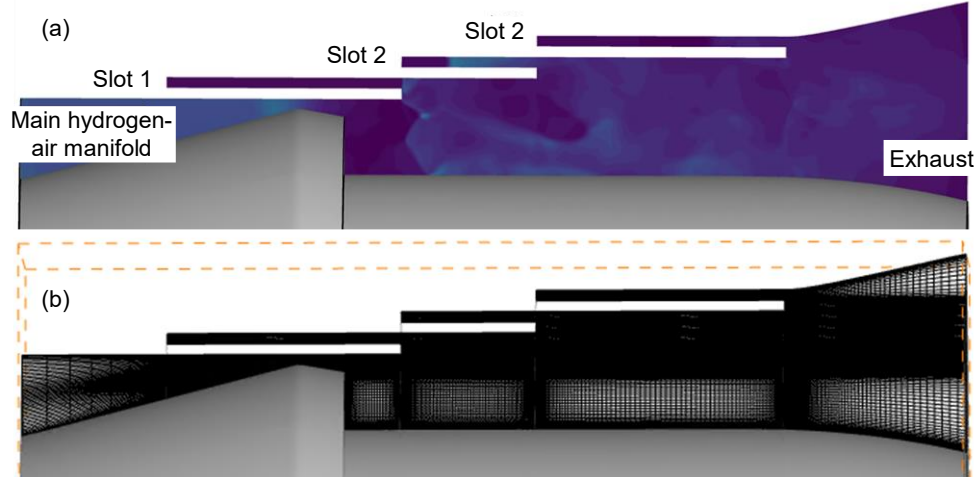


Figure 2: a) Cross-section of the numerical domain with three injection slots, b) structured mesh of the combustor.

After reaching steady state, Figure 3 shows a snapshot of unsteady interaction of the jets with the main detonation wave. At  $T = 1/8 T_{\text{detonation}}$ , the cooling slots do not have enough pressure to push cooling into the combustion chamber, but at  $T = 1/4 T_{\text{detonation}}$ , cooling starts to flow into the main combustion chamber all the way till  $T = 7/8 T_{\text{detonation}}$ .

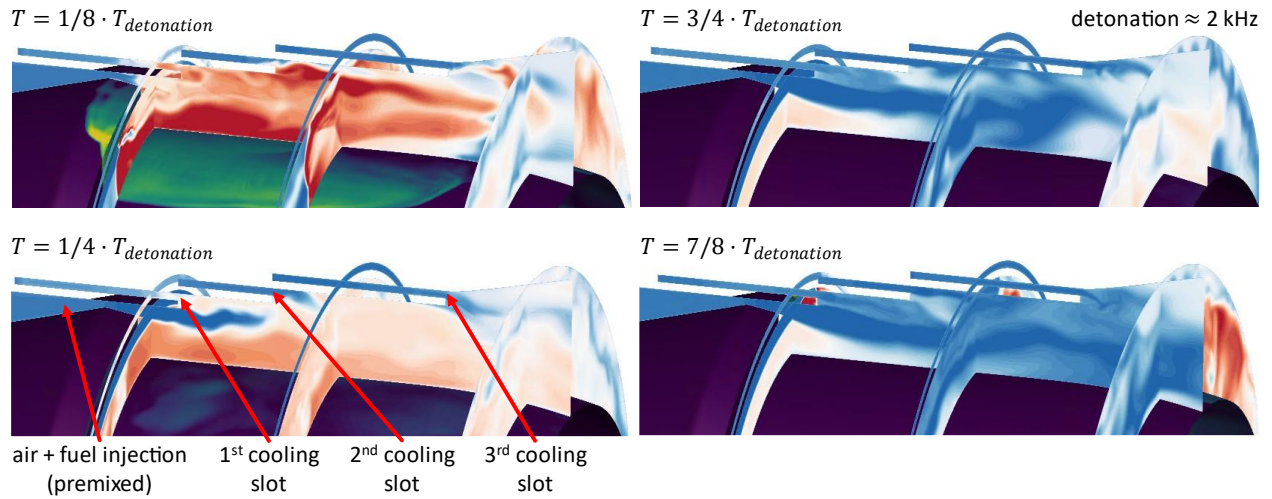


Figure 3: Four unsteady snapshots of temperature of the cooled combustor

Figure 4 shows the heat flux reduction thanks to the cooling scheme; a four-fold reduction was achieved with coolant injection on the shroud (upper end wall), whereas the inner end wall (hub) is unaltered because no cooling jacket was present.

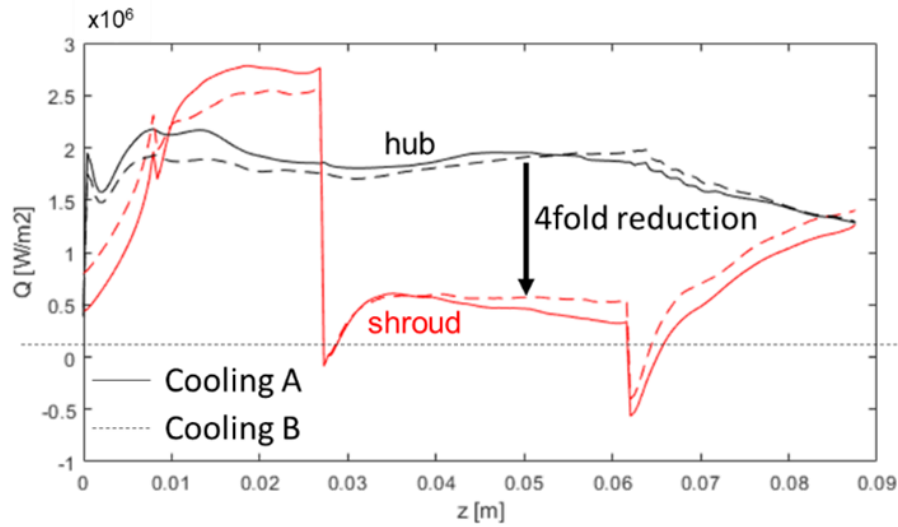


Figure 4: Heat flux along the axial direction: Comparison between cooled and uncooled combustor

The cooling mass flow constituted around 30% of the core mass flow. Additionally, the exhaust profile can be altered as shown in Figure 5; the uncooled combustor provides an exit temperature of 2000 K whereas the cooling jacket lowers the exit temperatures to around 1500 K on average while keeping a transonic-supersonic exhaust.

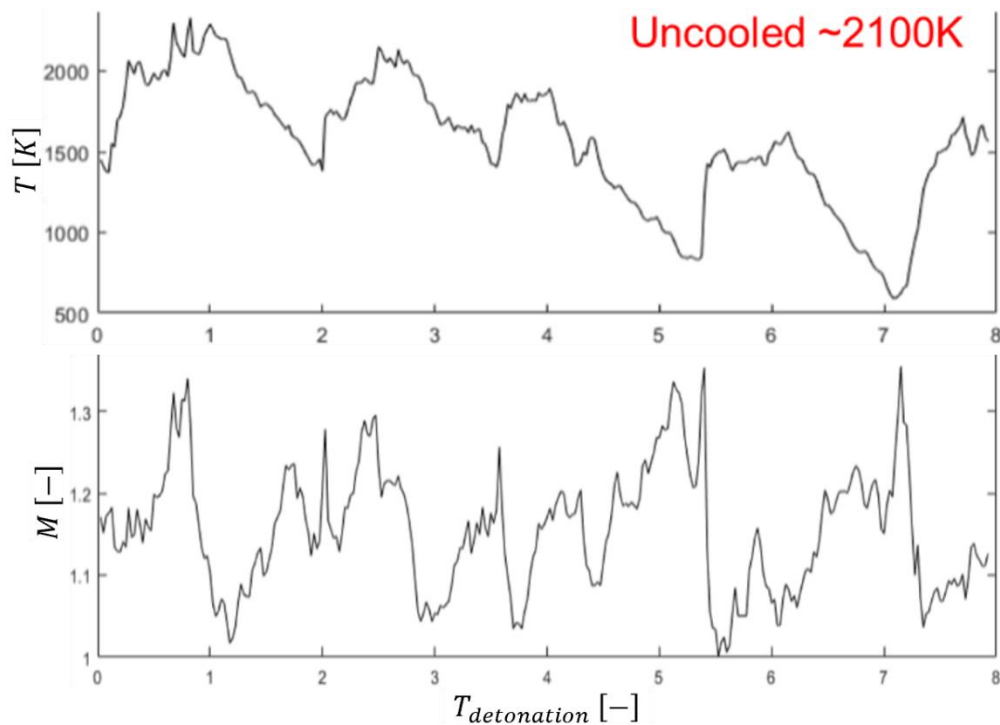


Figure 5: Mass flow averaged exhaust profile of temperature (top) and Mach number (bottom).

### 3. Design of the Combustor

#### 3.1. Design of the Combustor

The implementation of the cooling jacket in THOR's rotating detonation combustor is shown in Figure 6 below. In previous work, the university developed a modular RDC for optical diagnostics, so pieces of this RDC are reused for the cooled design[9]. The repurposed components include the combustion reactant air injection ring and the center body, which are shown in a slate gray color. The cooling jacket, shown in green, consists of the three previously described cooling slots as well as eight injection holes and 24 instrumentation ports. The eight injection holes are placed normal to the cooling jacket exterior, and two inner plenums redirect the cooling gas into the tangential cooling slots. The first plenum feeds the first two slots while the second plenum feeds the third slot. Each plenum is divided into four chambers around the cooling jacket circumference; each chamber occupies a  $90^\circ$  sector. Each plenum has four injection ports (one port per chamber). The two separate plenums allow for different injection parameters, such as gas composition, pressure, and/or temperature; this means that some experimental optimization may be conducted. Since the cooling sleeve has a complex geometry that would be difficult to machine, the cooling sleeve was designed to be 3D printed. Other design considerations specific to 3D printing are discussed in 4.1.

As discussed in 2.3, the CFD simulations predict that the RDC exit will have a rapidly fluctuating Mach number. However, the Mach number remains above unity, meaning that a diverging exit area will expand the flow, thereby increasing velocity and decreasing pressure. This diverging section is featured in the combustor flange, shown in pink in Figure 6. For this divergence, an angle of  $12.5^\circ$  to the horizontal was arbitrarily selected. The availability of mounting a turbine downstream of the detonation chamber, inside the diverging section of the flange was considered. The turbine design is not discussed in this paper, but the turbine was optimized for a low supersonic fluctuating flow. A unique miniature stator (Figure 7) was designed to straighten the combustor exit flow before it enters the turbine, to improve power extraction. The stator features a NACA 0030 blade profile, which indicates the max camber is 0% of the chord and the thickness is 30% of the chord. Even though the stator does not have distinct pressure and suction sides, the pressure and suction side nomenclature from the turbine are copied to the stator. Each stator blade is bolted to the combustor flange and has hub and tip profiles that match the center body and the divergence angle, respectively.

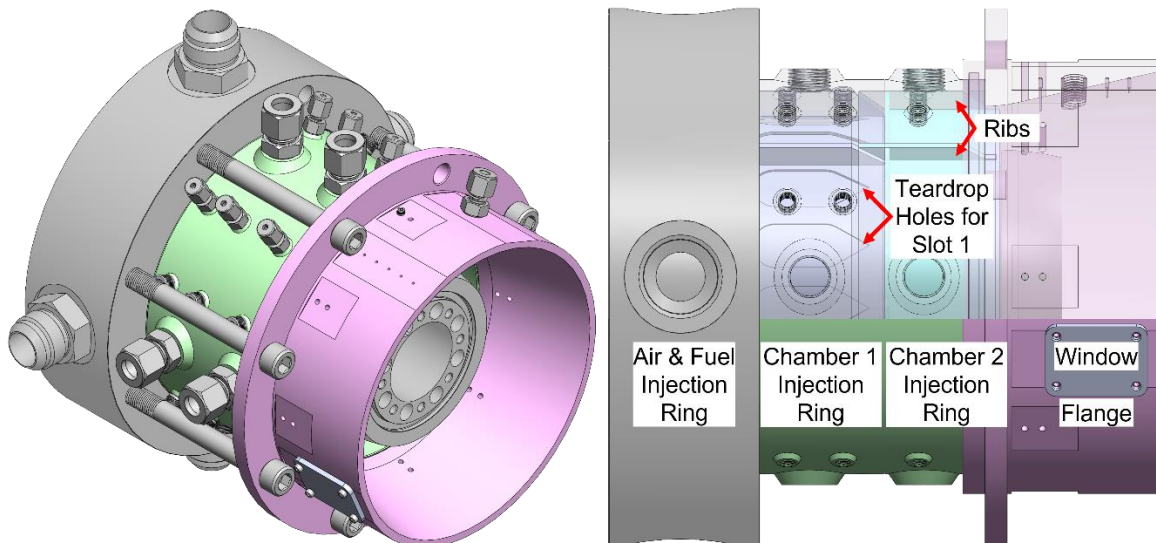


Figure 6: CAD of the cooled combustor.



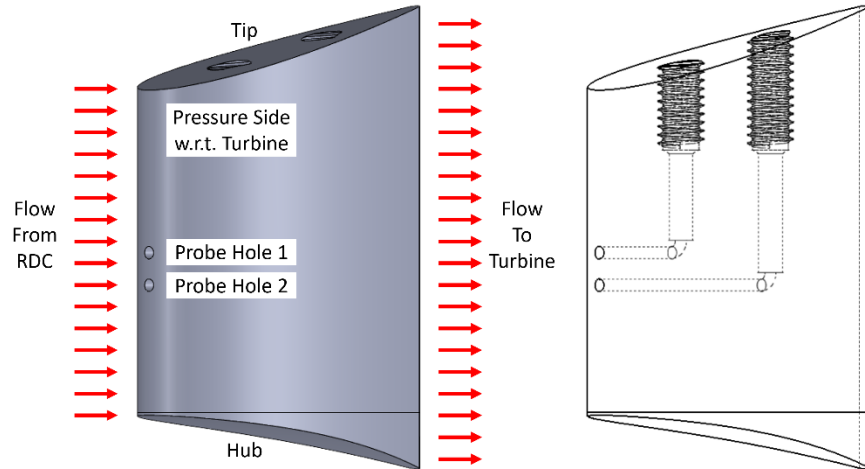


Figure 7: CAD of one stator blade.

### 3.2. Measurement Capabilities

As discussed in 2.2, it may be beneficial to choke the cooling slots since this would prevent backflow of the combustion products into the cooling chambers. To determine if the slots choke during the tests, pressure and temperature must be measured inside the cooling slots. To that end, the cooling jacket was designed with 24 threaded ports for instrumentation: sixteen for the first plenum and eight for the second plenum. These ports may be configured for pressure probes or thermocouples. Since the first plenum is shared by the first two cooling slots, care was taken to ensure that the pressures in the plenum and each slot may be measured separately. Ideally, the total pressure in the first plenum will match that of the first two slots, but the complex geometry may create pressure disturbances. Thermocouples may be placed in the plenums in a similar manner to the pressure probes, and surface thermocouples may also be utilized to determine the slot wall temperature.

In the diverging section of the flange, six static pressure taps are placed along the flow direction; these taps are spaced equally across the chord of the turbine blades. A small window overlooks the turbine blade flow path and may be used for optical access to the turbine. Although there are no probes inside the combustion chamber, the combustor exit is instrumented by implementing the stator blades. Each stator blade is mounted with two screws, and each screw hole is connected to 1/32" hole that serves as a pressure tap to the stator blade surface. These surface pressure taps are visible in Figure 7 and are labeled Probe Holes 1 and 2. Hollow screws (or vent screws) are used to connect the surface hole to an external pressure probe. Since there are eight flow straighteners, eight combinations of probe hole orientations were created. These combinations are listed in Table 1. As stated previously, the stators do not necessarily have pressure or suction sides since the stators have a 0° of incidence. However, for simplicity, the table below labels the stator sides with respect to the turbine pressure and suction sides.



Table 1: Flow straightener configurations.

| Stator Number | Probe Hole 1              |                    |                   | Probe Hole 2                 |                    |                   |
|---------------|---------------------------|--------------------|-------------------|------------------------------|--------------------|-------------------|
|               | Blade Side w.r.t. Turbine | Chord Position (%) | Span Position (%) | Side of Blade w.r.t. Turbine | Chord Position (%) | Span Position (%) |
| 1             | N/A                       | 0                  | 50                | N/A                          | 0                  | 40                |
| 2             | N/A                       | 0                  | 75                | N/A                          | 0                  | 25                |
| 3             | N/A                       | 100                | 40                | N/A                          | 100                | 50                |
| 4             | Suction Side              | 20                 | 50                | Pressure Side                | 20                 | 50                |
| 5             | Suction Side              | 40                 | 50                | Pressure Side                | 40                 | 50                |
| 6             | Suction Side              | 40                 | 30                | Suction Side                 | 40                 | 70                |
| 7             | Suction Side              | 60                 | 50                | Pressure Side                | 60                 | 50                |
| 8             | Suction Side              | 80                 | 50                | Pressure Side                | 80                 | 50                |

## 4. Experimental Assessment

### 4.1. Manufacturing of the Cooled Combustor

Finally, the cooling sleeve was designed to be made on a direct metal laser sinter 3D printer by Innovative 3D Manufacturing in Franklin, IN. The material was selected to be SAE 316L stainless steel since this offered fair corrosion resistance at high temperatures while also not being too expensive. Figure 8 shows the axial stepped cooling jacket, which was printed in the orientation shown in Figure 8b. One notable feature is the conical extension on each injection and instrumentation port. The cone angle was set to  $45^\circ$  such that no support material would need to be printed, thus reducing time on the printer and cost. Another feature (not shown) is that the wall separating Slots 1 and 2 has teardrop-shaped holes to allow cooling gas to pass from the first plenum into Slot 1. These teardrops terminate before injection into the detonation chamber. The angle of the teardrop was set to  $45^\circ$  to avoid support material, just like with the conical extensions. The teardrops did not necessarily save time on the printer, but there was no easy access to the plenum interior, hence the need for supportless printing. In addition to the cooling sleeve, a flange was designed to hold the cooling jacket to the base of the THOR rig. This sleeve was machined from carbon steel by Logo 2 Manufacturing in Alicante, Spain; carbon steel was chosen over other materials for its low cost and ease of machining. Figure 10a and b shows the flange at an intermediate stage and the final product, respectively; Figure 10c shows the optical access window insert (made from aluminum alloy 7075); and Figure 10d and e show the assembled combustor and exhaust flange.

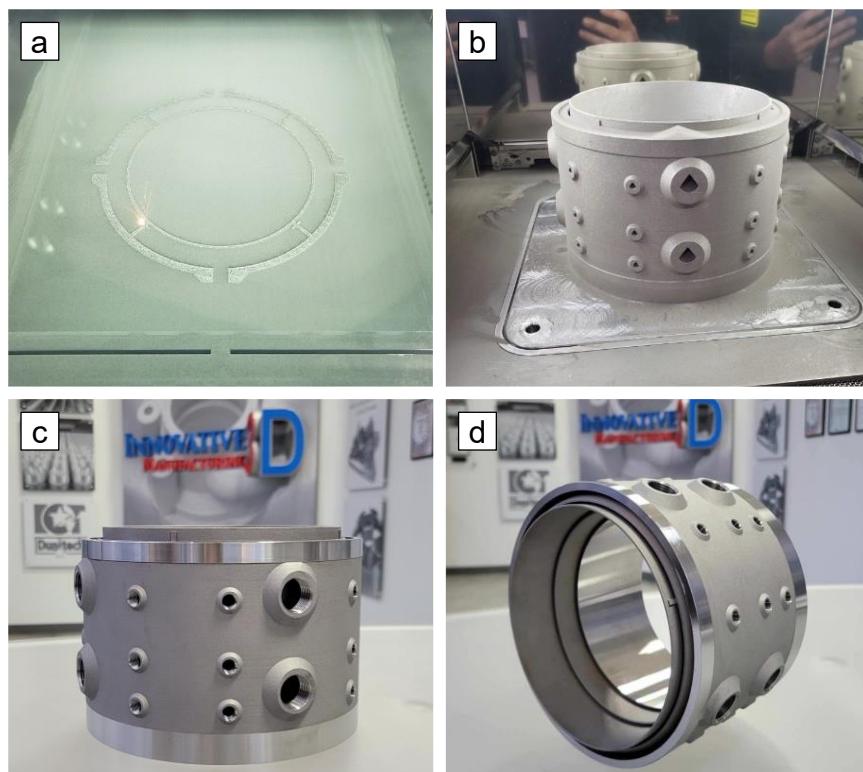


Figure 8: Manufacturing of the cooling jacket.

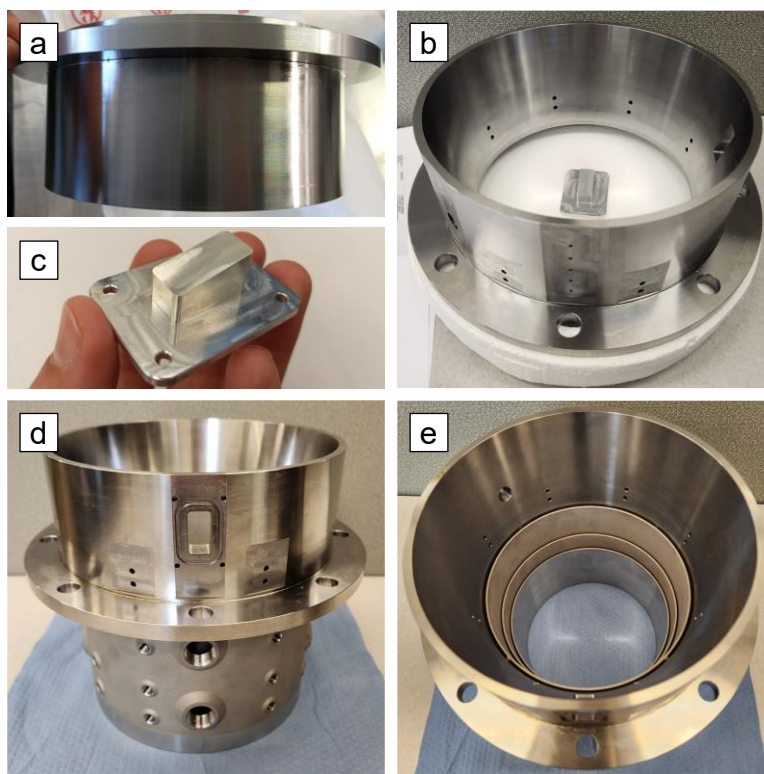


Figure 9: Manufacturing of the uncooled flange and window insert.

## 4.2. Test Data Analysis

Thermocouples were affixed to the rear-side of all three film-cooling slots to determine surface temperature of each cooling slot. Additionally, pressure taps were integrated into the film-cooling apparatus such that static pressures are also available for each slot. See the cutaway in Figure 10.

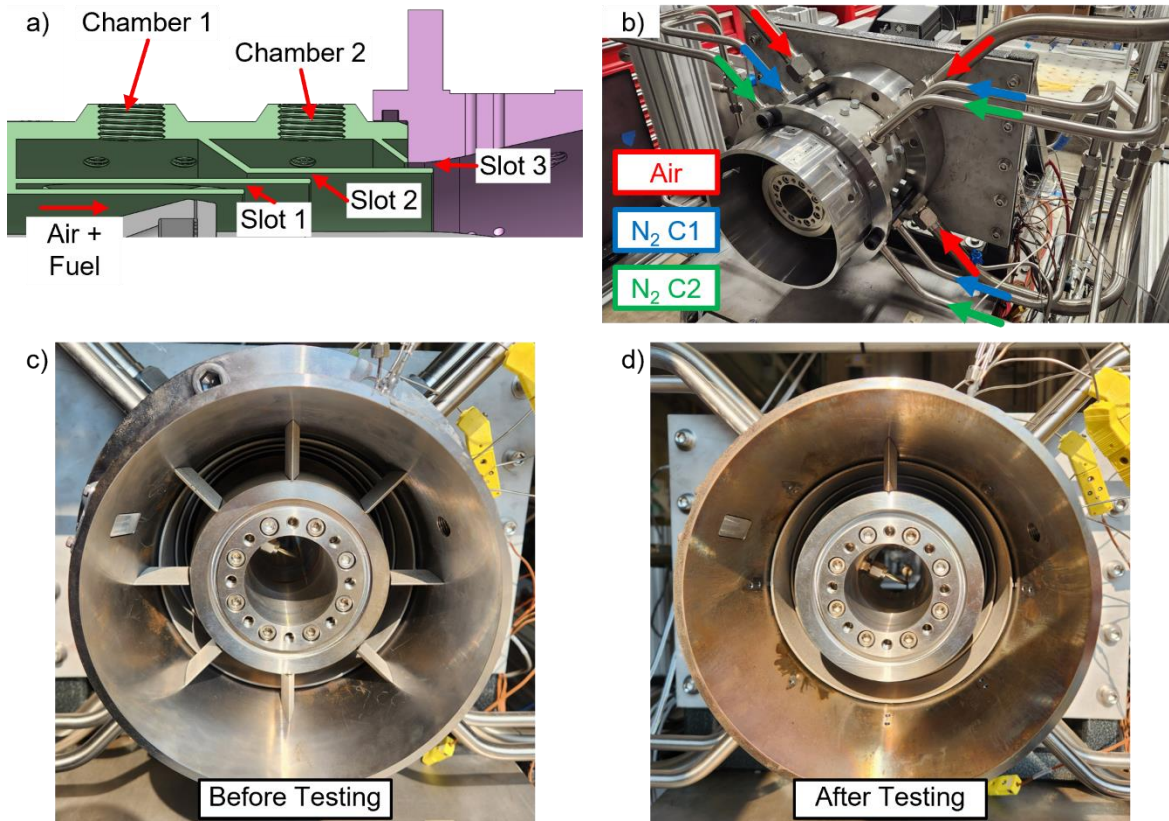


Figure 10 a) CAD cross section for instrumentation, b) layout in the THOR test stand, c) downstream imaging before and after the test

For all tests, a Phantom high-speed camera operating at 100 kHz was used in conjunction with a mirror to visualize the detonation waves from the aft of the combustor, looking forward. Due to the combustor exhaust during firing, the resulting high-speed footage exhibits noticeable distortion. Nevertheless, detonation waves were still visible and are shown in Figure 8. A still image taken before the test starts is shown on the left, and then the following sequence depicts a single detonation wave propagation in a clockwise direction. For clarity, the detonation wave is highlighted by the circle in the lower sequence.

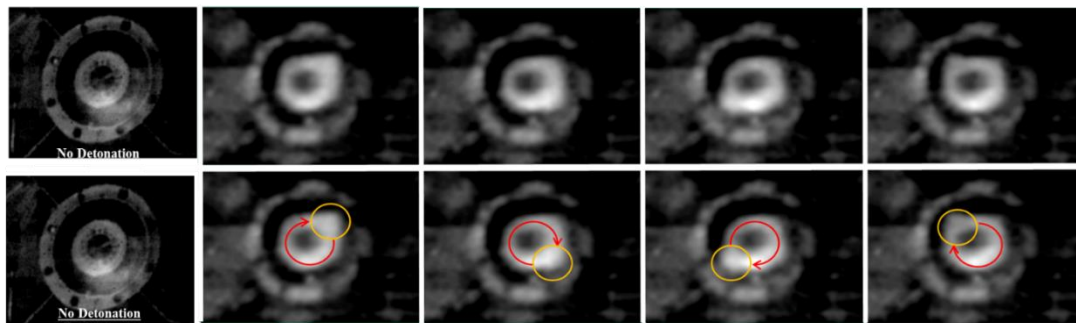


Figure 11 high speed imaging with single wave detection

The full test matrix for this campaign is shown in Table 2: Test matrix. Table 2. Despite 33 tests having been run, only 23 of the tests had successful ignition. The remaining were either cold-flows or failed-ignitions due to pre-det issues. For this data analysis, only the ignited tests have been considered. Test #32 is an uncooled control case.

Table 2: Test matrix.

| Test # | Duration (sec) | Type             | Air + H2 Mass Flow (lbm/s) | Air Pressure (psi) | H2 Pressure (psi) | Chamber 1 N2      |                | Chamber 2 N2      |                |
|--------|----------------|------------------|----------------------------|--------------------|-------------------|-------------------|----------------|-------------------|----------------|
|        |                |                  |                            |                    |                   | Mass Flow (lbm/s) | Pressure (psi) | Mass Flow (lbm/s) | Pressure (psi) |
| 1      | 0.3            | Cold             | 1                          | 334                | N/A               | 0.1               | 365            | 0                 | 0              |
| 2      | 0.3            | Cold             | 1                          | 334                | N/A               | 0                 | 0              | 0.09              | 370            |
| 3      | 0.3            | Cold             | 1                          | 334                | N/A               | 0.38              | 365            | 0.22              | 370            |
| 4      | 0.3            | Cold             | 1                          | 334                | N/A               | 0.29              | 365            | 0.19              | 370            |
| 5      | 0.3            | Ignition         | 1                          | 334                | 525               | 0.1               | 365            | 0.09              | 370            |
| 6      | 1              | No ignition      | 1                          | 334                | 525               | 0.1               | 365            | 0.09              | 370            |
| 7      | 1              | Ignition         | 1                          | 334                | 525               | 0.1               | 365            | 0.07              | 370            |
| 8      | 1.5            | No ignition      | 1                          | 334                | 525               | 0.1               | 365            | 0.1               | 370            |
| 9      | 2              | Partial ignition | 1                          | 334                | 525               | 0.1               | 365            | 0.1               | 370            |
| 10     | 2              | Ignition         | 1.39                       | 473                | 691               | 0.1               | 365            | 0.1               | 370            |
| 11     | 2.5            | No ignition      | 1                          | 333                | 527               | 0.1               | 365            | 0.1               | 370            |
| 12     | 2.5            | No ignition      | 1                          | 333                | 527               | 0.1               | 365            | 0.1               | 370            |
| 13     | 1              | Ignition         | 1                          | 341                | 499               | 0.1               | 365            | 0.1               | 380            |
| 14     | 2              | Ignition         | 1                          | 343                | 546               | 0.1               | 365            | 0.1               | 380            |
| 16     | 2              | Ignition         | 1                          | 340                | 540               | 0.1               | 364            | 0.1               | 369            |
| 17     | 2              | Ignition         | 1                          | 340                | 540               | 0.2               | 780            | 0.2               | 770            |
| 18     | 2              | Ignition         | 1                          | 340                | 540               | 0.2               | 730            | 0.2               | 745            |
| 19     | 2              | Ignition         | 1                          | 343                | 537               | 0.2               | 730            | 0.2               | 740            |
| 20     | 2              | Ignition         | 1                          | 342                | 532               | 0.3               | 1128           | 0.2               | 745            |
| 21     | 2              | Ignition         | 1                          | 341                | 642               | 0.3               | 1150           | 0.2               | 742            |
| 22     | 2              | Ignition         | 1                          | 343                | 643               | 0.3               | 1108           | 0.2               | 742            |
| 23     | 2              | Ignition         | 1.5                        | 490                | 756               | 0.3               | 1108           | 0.2               | 744            |
| 24     | 2.5            | Ignition         | 1.5                        | 490                | 823               | 0.3               | 1108           | 0.2               | 744            |
| 25     | 2.5            | No ignition      | 1.5                        | 506                | 805               | 0.3               | 1106           | 0.2               | 743            |
| 26     | 2.5            | Ignition         | 1.5                        | 504                | 806               | 0.3               | 1106           | 0.2               | 743            |
| 27     | 2.5            | Ignition         | 2                          | 655                | 1068              | 0.3               | 1110           | 0.2               | 739            |
| 28     | 3              | Ignition         | 2                          | 668                | 1070              | 0.3               | 1099           | 0.2               | 747            |
| 29     | 3              | Ignition         | 2                          | 666                | 1070              | 0.2               | 730            | 0.2               | 742            |
| 30     | 3              | Ignition         | 2                          | 671                | 1066              | 0.3               | 1096           | 0.3               | 1137           |
| 31     | 3              | Ignition         | 2                          | 669                | 1063              | 0.4               | 1458           | 0.3               | 1107           |
| 32     | 0.5            | Ignition         | 1                          | 344                | 539               | 0                 | 0              | 0                 | 0              |
| 33     | 5              | Ignition         | 1                          | 340                | 548               | 0.3               | 1106           | 0.2               | 738            |

As an initial demonstration of the cooling-slot effectiveness, the temperature profile of the uncooled control case (Test #32) is plotted alongside that of Test #33, the longest duration cooled test; these temperature profiles are depicted in Figure 12. The vertical dashed lines denote the start and end times of each test. As can be seen, the uncooled test experiences a rapid increase in slot temperature over the short test duration, whereas the cooled test experiences no significant increase in slot temperature, enabling a much longer test duration.

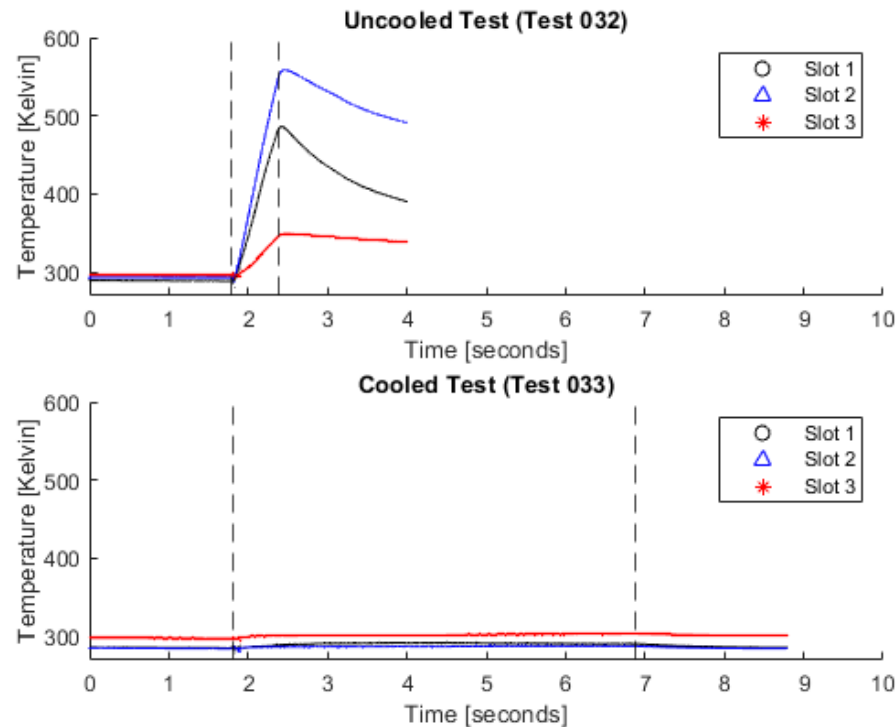


Figure 12: Temperature vs. time for an uncooled (top) and a cooled test (bottom).

The start and end times used here are derived from thrust measurements by virtue of a load-cell integrated into the RDC frame. Due to installation, an absolute value of the RDC thrust was not measured, however the load-cell data provides a relative comparison that enables the determination of the test window, as shown in Figure 13. A rolling-average is applied to smooth the noisy thrust data, and a threshold value is set halfway between the initial sensor value and the maximum recorded value. The test duration is then simply found by counting the time spent above the threshold.

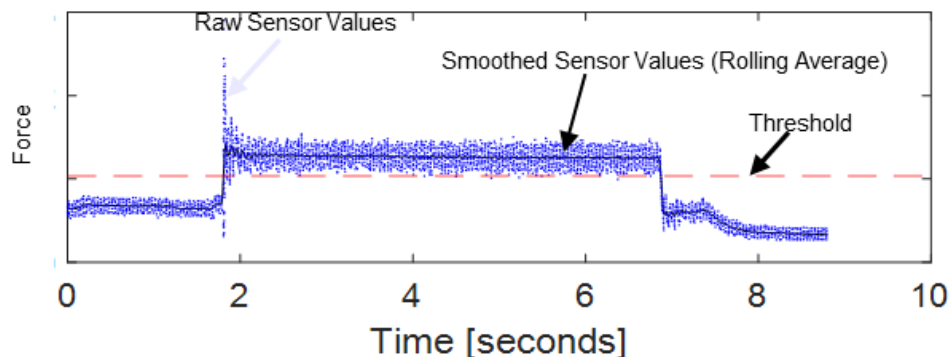


Figure 13: Start-stop signal for the test based on load cell information.



This simple method was found to be robust enough to determine test start and end points for all ignited tests. Figure 14: Test duration for each test shows the measured test-time from this method, which can be compared to the target times in the test matrix shown in Table 2.

```

Test Duration for Test005 : 0.255 seconds
Test Duration for Test007 : 0.940 seconds
Test Duration for Test010 : 1.006 seconds
Test Duration for Test013 : 1.081 seconds
Test Duration for Test014 : 2.078 seconds
Test Duration for Test015 : 2.078 seconds
Test Duration for Test016 : 2.077 seconds
Test Duration for Test017 : 2.076 seconds
Test Duration for Test018 : 2.076 seconds
Test Duration for Test019 : 2.078 seconds
Test Duration for Test020 : 2.076 seconds
Test Duration for Test021 : 2.074 seconds
Test Duration for Test022 : 2.078 seconds
Test Duration for Test023 : 2.075 seconds
Test Duration for Test024 : 2.570 seconds
Test Duration for Test026 : 2.569 seconds
Test Duration for Test027 : 2.577 seconds
Test Duration for Test028 : 2.578 seconds
Test Duration for Test029 : 2.580 seconds
Test Duration for Test030 : 2.576 seconds
Test Duration for Test031 : 2.574 seconds
Test Duration for Test032 : 0.587 seconds
Test Duration for Test033 : 5.066 seconds

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Figure 14: Test duration for each test.

To determine the impact of the coolant mass-flow on the cooling performance, the blowing ratio (BR) is defined as follows:

$$BR = \frac{\dot{m}_{coolant}}{\dot{m}_{fuel+oxidizer}} = \frac{\dot{m}_{N_2}}{\dot{m}_{air} + \dot{m}_{H_2}} \quad (4)$$

A higher value of BR indicates more cooling; for instance, a value of 1 indicates that the cooling mass flow rate is equal to propellant mass flow rate, whereas a value of 0 indicates an uncooled test. For these experiments, the blowing ratio never exceeded 0.5 to avoid ignition/detonation difficulty due to excessive nitrogen dilution.

The slot temperature increase can be shown by the average rate of slot temperature change, expressed as the following ratio:

$$\frac{(T_{final} - T_{initial})}{t_{test}} \quad (5)$$

where  $T_{final}$  and  $T_{initial}$  are the temperatures recorded at the end and at the start of the test respectively, and  $t_{test}$  is the test duration. The decision to use this rate instead of a direct temperature delta ( $T_{final} - T_{initial}$ ) or ratio ( $T_{final}/T_{initial}$ ) was made to remove the dependence on test duration.

By plotting this rate against the previously defined blowing ratio, a correlation between coolant mass flow rate and slot temperature increase rate can be observed (Figure 15), colored by the total mass flow rate (coolant and

propellant). During the testing, high-speed camera footage revealed that in many tests, a single detonation wave was not observed. Rather, two or more counter-rotating waves were seen in a “slapping mode”. To indicate these non-singular modes, the unfilled markers in Figure 15 correspond to multi-wave solutions, whereas the single-wave solutions are shown as filled-in markers.

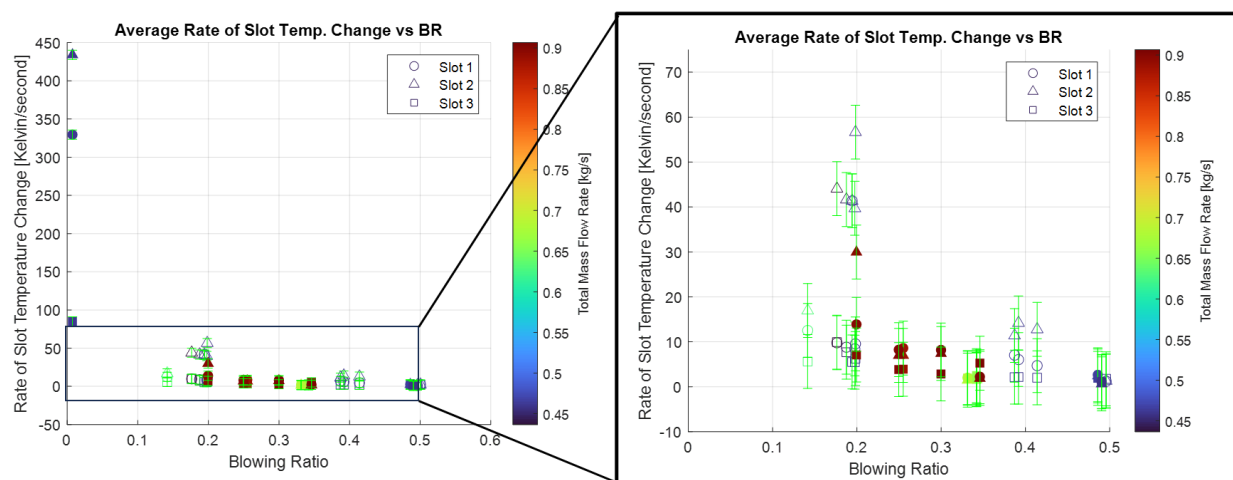


Figure 15: Average rate of slot temperature change.

As can be seen, the uncooled test has the highest slot temperature change, but as cooling increases, the slot temperature change is less pronounced. For a clearer view, the slapping-mode tests have been omitted in Figure 16. Uncertainty bands are based on a thermocouple uncertainty of  $\pm 3$  Kelvin, however conduction and velocity errors are unaccounted for in this analysis. Above blowing ratios of 0.2, the rate of change in the slot temperature was below 10 K/second. Slot 2 featured the highest rate of change, possibly due to the highest strength of the detonation wave at that location.

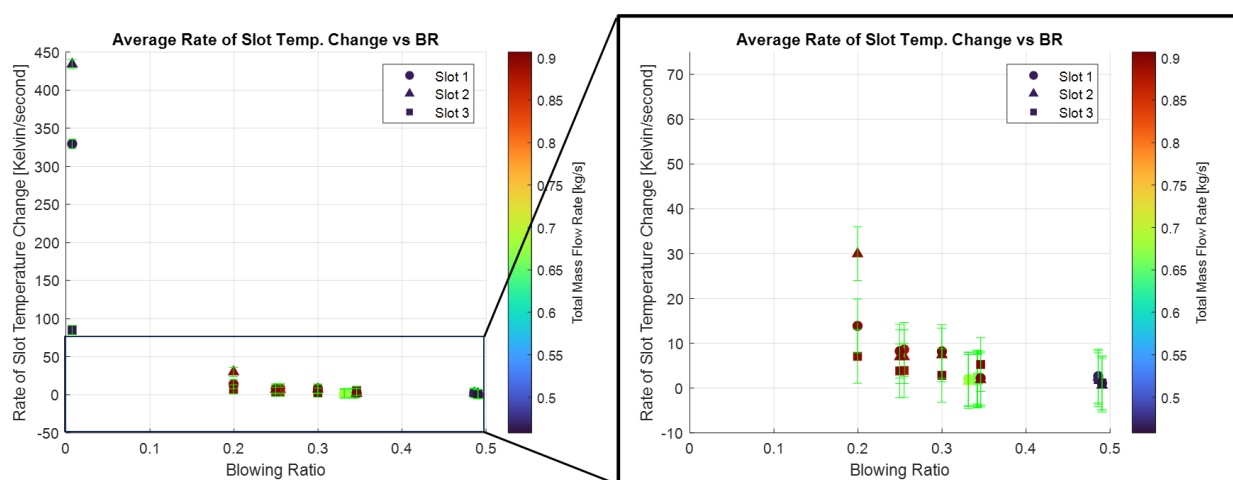


Figure 16: Average rate of slot temperature change for single-wave-mode tests.

Slot pressures were also plotted for comparison between the cooled and uncooled tests. The feed slots observed a higher-pressure during combustion because the slots experienced back flow from the RDE as the detonation wave passed each slot. This is visible in Figure 17 (bottom), for which the pressure increases during the combustion wave.



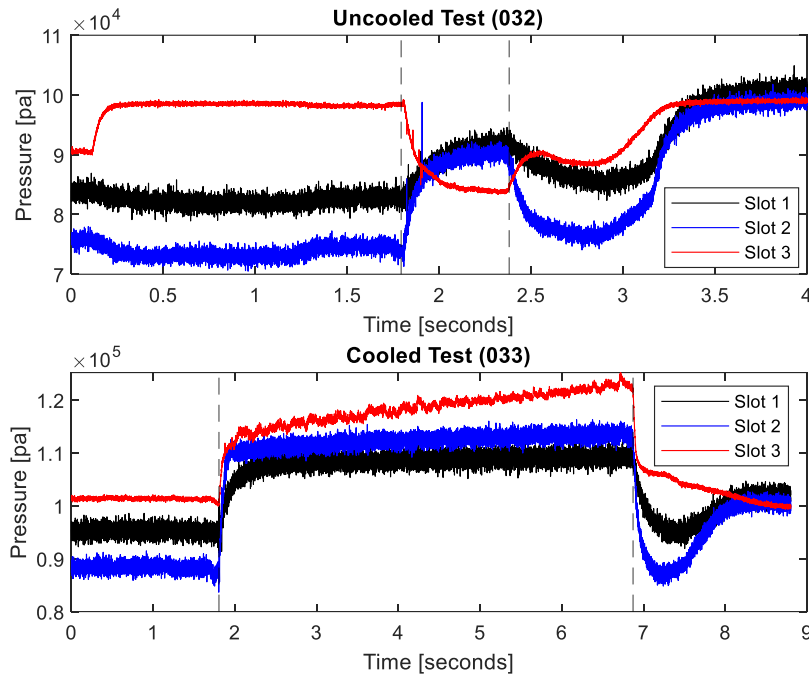


Figure 17: Pressure trace as a function of time for an uncooled test (top) and cooled test (bottom).

The slot pressure ratios (final vs. initial) are plotted against the blowing ratio in Figure 18. Once again, single wave solutions are denoted with solid markers, and hollow markers indicate multi-wave solutions. These results are also colored by the slot temperature change. In average slot 2 experiences the highest blockage from the wave, and single wave detonation events influence the cooling plenums more. The third cooling slot was less affect by the detonation wave as the wave strength was already decreased. For a blowing ratio near 0.3, slot 2 featured highest pressure increase during the hot fire.

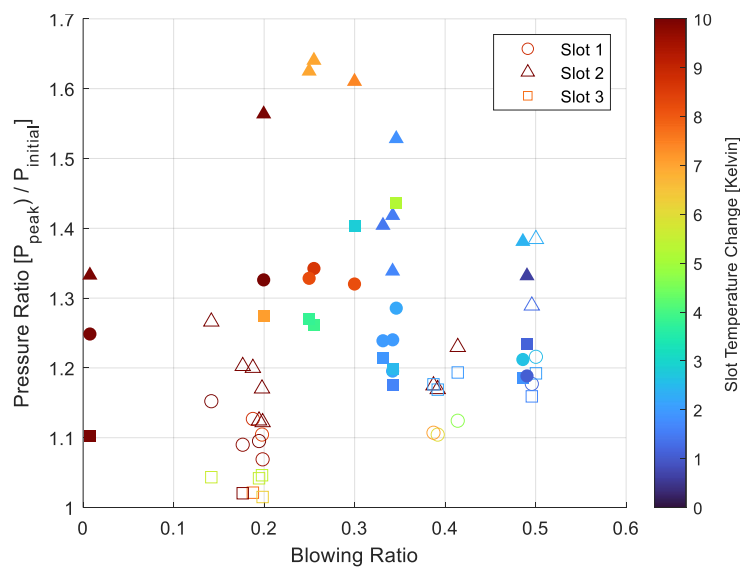


Figure 18: Slot static pressure versus blow ratio.

## 5. Conclusions

A new cooling scheme for rotating detonation combustors was successfully demonstrated in this paper. The stepped cooling scheme was first analyzed starting from Goldstein's work and numerically assessed through CFD on a rotating detonation combustor. This rotating detonation combustor was already tested in an uncooled configuration, and a cooling jacket was designed with CAD and created by 3D printing. Finally, an experimental campaign was carried out on an existing RDE test platform with the axial injection stepped cooling slots. High speed imaging and pressure and temperature probes were used to quantify the amount of cooling that was added and the effectiveness of the cooling scheme. The temperature readings recorded successful cooling of the hardware throughout the test matrix at low and high blowing ratios. The experiments revealed a wide range of operating conditions for which a single mode was detected, but for various conditions of the cooling slot, other modes were also detected such as the slapping mode operation. Above blowing ratios of 0.2, the rate of change in the three slots was minimal. The effect of the wave on the pressure within the cooling jacket was also assessed, and pressure blockage of up to 60% was found in the middle slot (which also featured highest temperature changes), whereas the downstream slot was less affected by the detonation wave.

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